

# OPERATING SYSTEMS

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## Process Management and Scheduling

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## Process Management Commands

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- **ps** – to view processes on the system
- **ps -f** – to view process with additional attributes
- **ps -aux** –all users processes
- **ps -al** –lists all active processes of a user
- **pstree** – visualize the process in a tree format
- **top** - to monitor the system as it shows the system resources used by different processes. It gives a snapshot of the situation that the system is currently in.

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- **htop** – It is an interactive text mode process viewer. It displays the per CPU usage and memory, swap usage using a text graph.
- we can use the Up/Down arrow key to select processes, F7 and F8 to change the priority and F9 to kill a process.



- **Nice** - set the priorities of new processes By default, a process gets launched with priority 0.
- Process priority can be viewed using the top command output under the NI (nice value) column.
- Values of process priority range from -20 to 19. Lower the nice value, higher the priority.
- Syntax : nice <priority value> <process name>

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- **kill** - terminate processes
- **kill -9** – forcefully kill
- Syntax: `kill -9 <pid>`
- **pkill** - kill processes using a pattern, i.e. process name, process owner etc
- Syntax: `pkill <process name>`



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- **&** - To start a process in background
- Eg: `find . -name *iso > /tmp/res.txt &`
- **jobs** – Lists all the background jobs
- Foreground process to background
  - CTRL+Z' and 'bg' command
- background process to foreground
  - `fg %<no>`



- Start/stop/restart Deamon processes
  - `service <servicename> stop`
  - `service <servicename> start`
  - `service <servicename> restart`
  - Eg: `service apache2 start`
- `service - -status-all` - list all the daemons

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- **Renice** – change the priority of the already executing process
- Eg: `renice -4 -p <pid>`



- `ipcs` - listing the inter-process communication facilities
- `ipcs -q` – List all message queues
- `ipcs -s` – List all semaphores
- `ipcs -m` – List all shared memory
- `ipcs -q -i <pid>` - option provides detailed information about `q` ipc facility for that process
- `ipcs -q -l` - gives the system limits for `q` ipc facility.
- `ipcs -p` – processid which accessed the ipc facility recently

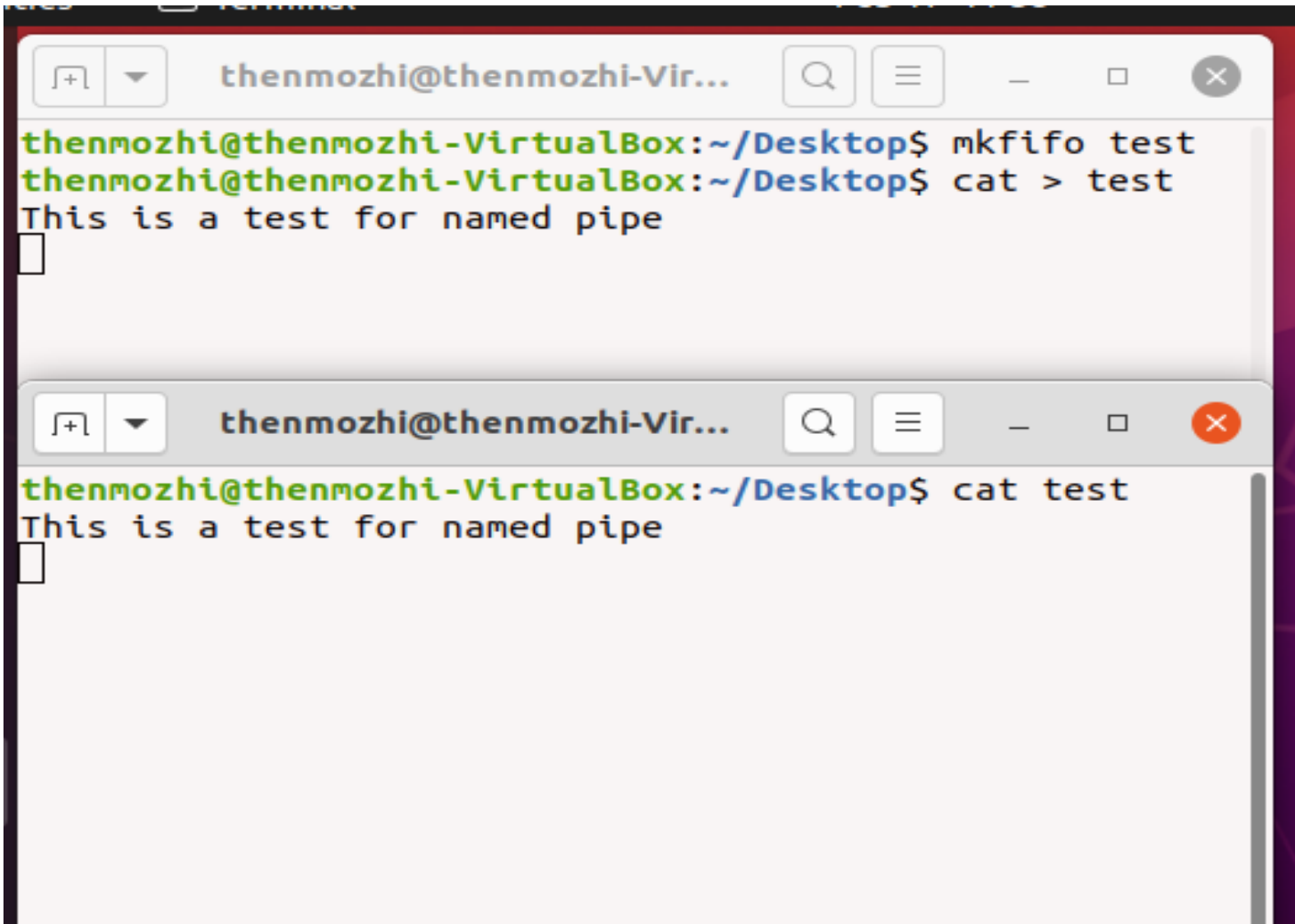
- | - represent a unnamed pipes. The first command is one process and the second command is another process, both of them do a communication through unnamed pipes. The output of the first command goes as an input to the next command.
  - Eg: `cat one.txt | grep "test"`

- We can create a named pipe by creating a fifo file using mkfifo command.
  - Eg: Mkfifo mypipe
- This creates a pipe file and contents are written to this file only when the process executes. When the process finishes its execution, the file contents will be cleared and made as an empty file.

- To test the contents written into the fifo file, open two terminals
- In one terminal `cat > mypipe` and in another terminal put `cat mypipe` and start inputting the text in the file. Now you can see the contents in the other terminal. But once you close the file in one terminal, the content is cleared from the file.

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## Named Pipes



```
thenmozhi@thenmozhi-VirtualBox:~/Desktop$ mkfifo test
thenmozhi@thenmozhi-VirtualBox:~/Desktop$ cat > test
This is a test for named pipe
^C

thenmozhi@thenmozhi-VirtualBox:~/Desktop$ cat test
This is a test for named pipe
^C
```



# THANK YOU

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